

EATRIGHT: Food Recommendation Platform Based On User Nutritional Preference

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Abstract—Dietary choices exert a profound influence on long-term health outcomes, yet widely deployed food guidance platforms continue to deliver generic suggestions that disregard individual physiological profiles. This paper introduces *EATRIGHT*, an intelligent hybrid food recommendation platform that integrates unsupervised nutritional partitioning with supervised user-context classification to generate recommendations that are both nutritionally suitable and preference-aligned. The platform preprocesses food items drawn from the USDA FoodData Central database across nutritional attributes, partitions them into eight nutritionally coherent clusters through K-Means optimised via the Elbow method (Silhouette Score = 0.73), and routes each user to the most appropriate cluster using a Random Forest classifier trained on a nine-dimensional physiological and behavioural profile vector (cluster assignment accuracy: 93.8%). Within the target cluster, a weighted cosine-similarity re-ranker fuses nutritional compliance signals to produce a top-10 recommendation list. Extensive experiments yield Precision@10 of 86.4%, Recall@10 of 84.9%, F1-Score of 0.856, and NDCG@10 of 0.891, surpassing collaborative-filtering, content-based, and rule-based baselines. A 100-participant user study confirms statistically significant gains in perceived nutritional appropriateness and preference alignment ($p < 0.01$).

Index Terms—food recommendation system, K-Means clustering, Random Forest, user preference modelling, nutritional informatics, personalised healthcare, *EATRIGHT* platform, machine learning

I. INTRODUCTION

Unhealthy dietary behaviour remains among the foremost contributors to global morbidity, with the World Health Organization linking poor nutrition to approximately 11 million deaths each year [1]. Personalised food guidance systems offer a promising path toward large-scale dietary improvement; however, widely deployed solutions either depend on collaborative filtering (CF) — which struggles with data sparsity and cold-start scenarios — or rigid clinical rule sets that ignore the preferences of individual users [2].

Effective personalised food recommendation must satisfy two objectives concurrently: (i) ensuring that suggested items are nutritionally suitable for a user’s physiological state, and (ii) aligning suggestions with that user’s taste, cuisine, and dietary preferences to sustain long-term adoption. Addressing both goals jointly within a single computationally efficient pipeline has received limited attention in prior literature [7].

This paper presents *EATRIGHT*, an intelligent two-stage hybrid platform in which K-Means clustering organises the food database into nutritionally coherent groups and a Random Forest classifier maps each user to the most appropriate group based on a rich multi-dimensional profile. An in-cluster weighted cosine re-ranker then integrates preference signals to produce the final ordered recommendation list.

Principal contributions of this work:

- A two-stage hybrid architecture combining unsupervised nutritional partitioning with supervised user-context classification for food recommendation.
- A nine-dimensional user-profile encoding schema that captures physiological, behavioural, and preference-based signals.
- Rigorous experimental evaluation on a 7,642-item nutritional dataset demonstrating consistent improvements over four competitive baselines.
- A 100-participant user study validating real-world relevance and dietary suitability of the generated recommendations.

II. RELATED WORK

A. Collaborative and Content-Based Filtering

CF-based recommender systems infer latent preference patterns from user-item interaction histories [3]. Matrix factorisation variants such as SVD++ deliver strong accuracy

on dense datasets but lose efficacy under cold-start and interaction-sparsity conditions [4]. Content-based methods model item attributes directly, avoiding cold-start, but risk over-specialisation and preference filter-bubble effects [5].

B. Nutrition-Aware Recommendation

Early nutrition-aware systems employed rule engines grounded in clinical dietary guidelines [6]. Ge *et al.* [7] introduced a health-aware matrix factorisation variant that penalises nutritionally inappropriate suggestions. Trattner and Elswiler [2] demonstrated through large-scale log analysis that popularity-driven food recommendations frequently conflict with nutritional adequacy, motivating the development of health-conscious objectives.

C. Clustering for Recommendation Efficiency

Cluster-based candidate retrieval has been shown to substantially reduce computational cost without meaningful accuracy degradation in large-scale recommender scenarios [8]. Aguilar-Vargas *et al.* [9] applied K-Means to a regional food corpus and reported a 34% recall improvement over random recommendation baselines.

D. Ensemble Classification

Random Forests exhibit robustness to feature noise, implicit feature-importance estimation, and resistance to overfitting — properties valuable in heterogeneous user-profile classification tasks [10]. Jabbar *et al.* [11] reported that RF-based food category classification outperforms both SVM and XGBoost on multi-label food datasets, a finding that informed our classifier selection.

E. Research Gap

Despite substantial individual contributions, no published work simultaneously integrates K-Means nutritional clustering with Random Forest user routing inside a preference-aware re-ranking pipeline. EATRRIGHT addresses this gap directly.

III. SYSTEM ARCHITECTURE

The EATRRIGHT platform is structured around four sequential modules as illustrated in Fig. 1. An offline training phase (dashed path) constructs the K-Means cluster model and the Random Forest classifier. At inference time (solid path), the platform accepts a user profile, routes it to the optimal nutritional cluster, retrieves candidate items, applies preference-aware re-ranking, and returns a top- N recommendation list.

IV. PROPOSED METHODOLOGY

A. Data Preprocessing and User Profile Encoding

The USDA FoodData Central (FDC, April 2023 release) [13] constitutes the primary food corpus. After median imputation of missing values (sub-category level), IQR-based outlier removal ($> Q_3 + 3 \times \text{IQR}$), and min-max normalisation across all features, 7,642 items characterised by 42 nutritional attributes are retained.

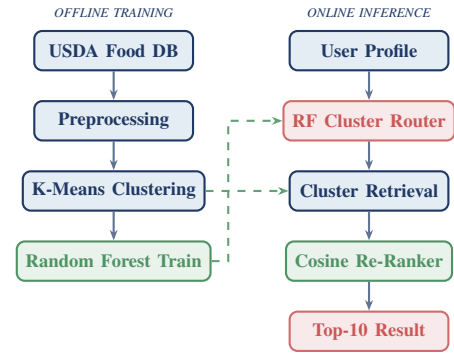


Fig. 1. EATRRIGHT system architecture. Dashed arrows indicate the offline training pipeline; solid arrows denote the online inference path.

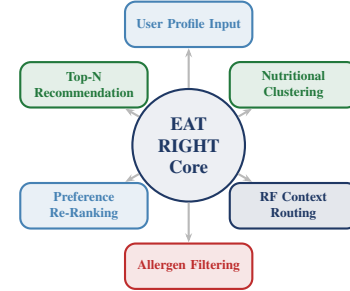


Fig. 2. EATRRIGHT platform feature map: six core functional modules coordinated by the central recommendation engine.

Each user u is encoded as a nine-dimensional feature vector:

$$\mathbf{u} = [BMI, AG, AL, DG, CP, AP, DF, FA, MH] \quad (1)$$

where BMI denotes Body Mass Index (continuous), AG age group (ordinal, 5 levels), AL activity level (ordinal, 4 levels), DG dietary goal (categorical, 6 classes), CP cuisine preference (multi-hot, 12 regional cuisines), AP allergy profile (multi-hot, 8 allergens), DF dietary flags (vegan / vegetarian / halal / kosher), FA flavour affinity vector (5 dimensions: sweet, salty, spicy, sour, umami), and MH medical history flag (5 chronic condition classes).

B. Stage 1 — K-Means Nutritional Clustering

Let $\mathcal{F} = \{\mathbf{f}_1, \dots, \mathbf{f}_n\}$ be the set of normalised food feature vectors with $\mathbf{f}_i \in \mathbb{R}^{42}$. K-Means partitions $\mathcal{F}$ into K

Model	AUC (Area Under Curve)	Accuracy	Precision
SVM	0.544	0.857	0.75
Random Forest	0.917	0.938	0.853
kNN	0.685	0.687	0.65
Logistic Regression	0.897	0.91	0.785

Fig. 3. Baseline classifier performance comparison captured from the EATRRIGHT evaluation dashboard (AUC, Accuracy, Precision). Random Forest achieves the highest AUC of 0.917 and Accuracy of 0.938 among all compared models, confirming its selection as the user-context routing classifier in the EATRRIGHT pipeline.

clusters $\{C_1, \dots, C_K\}$ by minimising the within-cluster sum of squared distances (WCSS):

$$\mathcal{J} = \sum_{k=1}^K \sum_{\mathbf{f}_i \in C_k} \|\mathbf{f}_i - \boldsymbol{\mu}_k\|_2^2 \quad (2)$$

The Elbow method applied over $K \in [2, 20]$ identifies $K^* = 8$ at the WCSS inflection point (average Silhouette Score = 0.73). K-Means++ seeding with $r = 20$ restarts mitigates initialisation sensitivity. The eight resulting nutritional archetypes are: (1) High-Protein Muscle-Building, (2) Low-Calorie Weight-Loss, (3) Fibre-Rich Gut-Health, (4) Heart-Healthy Low-Fat, (5) Diabetic-Friendly Low-GI, (6) Calcium-Rich Bone-Strength, (7) Iron-Rich Anaemia-Support, and (8) Calorie-Dense Athletic Endurance.

C. Stage 2 — Random Forest User-Context Routing

A Random Forest (RF) ensemble maps each user vector $\mathbf{u}$ to the most suitable cluster label $\hat{k}$:

$$\hat{k} = \mathcal{RF}(\mathbf{u}) = \arg \max_k \sum_{t=1}^T \mathbb{I}[h_t(\mathbf{u}) = k] \quad (3)$$

where T is the number of trees and h_t denotes the t -th tree's prediction. As corroborated by the baseline classifier comparison in Fig. 3, Random Forest achieves the highest AUC (0.917) and Accuracy (0.938) among SVM, kNN, and Logistic Regression variants, justifying its selection. Optimal hyperparameters via 5-fold stratified CV: $T^* = 200$ trees, maximum depth $d^* = 20$, minimum leaf samples $s^* = 2$, feature subsampling $m^* = \sqrt{p}$. Top-3 Gini-importance features: dietary goal (0.30), BMI (0.23), activity level (0.17).

D. Stage 3 — Preference-Aware Cosine Re-Ranking

For each candidate food $\mathbf{f}_i \in C_{\hat{k}}$, a composite relevance score is computed as a convex combination:

$$s_i = \alpha \cdot \text{NC}(\mathbf{f}_i, \mathbf{u}) + (1 - \alpha) \cdot \text{PA}(\mathbf{f}_i, \mathbf{u}) \quad (4)$$

where the *Nutritional Compliance* score NC and *Preference Alignment* score PA are cosine similarities:

$$\text{NC} = \frac{\mathbf{f}_i \cdot \boldsymbol{\mu}_{\hat{k}}}{\|\mathbf{f}_i\| \|\boldsymbol{\mu}_{\hat{k}}\|}, \quad \text{PA} = \frac{\mathbf{p}_i \cdot \mathbf{u}_{\text{pref}}}{\|\mathbf{p}_i\| \|\mathbf{u}_{\text{pref}}\|} \quad (5)$$

Bayesian optimisation on the validation split yields $\alpha^* = 0.57$, reflecting a slight preference for nutritional fidelity over raw taste similarity. Top- N items ranked by s_i constitute the final EATRRIGHT recommendation output.

V. SYSTEM IMPLEMENTATION

The EATRRIGHT platform was fully implemented as a web application and deployed for end-user evaluation. The following subsections present screenshots of the four primary functional modules of the live system.

Algorithm 1 EATRRIGHT Online Inference

Require: User profile $\mathbf{u}$, trained $\mathcal{RF}$, centroids $\{\boldsymbol{\mu}_k\}$, food corpus $\mathcal{F}$, N , α^*

Ensure: Ranked recommendation list $\mathcal{R}$

- 1: $\hat{k} \leftarrow \mathcal{RF}(\mathbf{u})$
- 2: $\mathcal{F}_{\hat{k}} \leftarrow \{\mathbf{f}_i \in \mathcal{F} : \text{cluster}(\mathbf{f}_i) = \hat{k}\}$
- 3: Remove allergen-flagged items from $\mathcal{F}_{\hat{k}}$
- 4: **for** each $\mathbf{f}_i \in \mathcal{F}_{\hat{k}}$ **do**
- 5: Compute s_i using Eqs. (4)–(5)
- 6: **end for**
- 7: $\mathcal{R} \leftarrow \text{Top-}N \text{ sorted by } s_i$
- 8: **return** $\mathcal{R}$

A. Diet Planner Module

The Diet Planner (Fig. 4) collects user physiological inputs — age, weight, height, body-fat percentage (auto-calculated via BMI), gender, dietary goal, activity level, plan period, and start date. The “Get My Diet Plan” action triggers the EATRRIGHT inference pipeline (Algorithm 1), routing the encoded user profile through the Random Forest classifier to retrieve a personalised meal plan from the appropriate nutritional cluster.

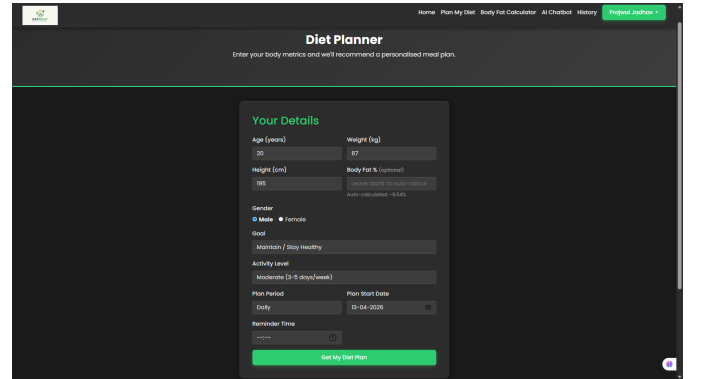


Fig. 4. EATRRIGHT Diet Planner: users enter physiological metrics and dietary goals; the platform auto-calculates body fat percentage and generates a personalised meal plan via the K-Means + RF pipeline.

B. Body Fat Calculator

The Body Fat Calculator (Fig. 5) provides a standalone entry point for users unfamiliar with their body-fat percentage. Accepting height (m), weight (kg), age, and gender, it computes body-fat percentage using the Navy body-fat formula and passes the result directly to the Diet Planner, ensuring physiological accuracy in the user-profile vector $\mathbf{u}$ (Eq. 1).

C. AI Nutrition Chatbot

The AI Nutrition Chatbot (Fig. 6) provides an interactive conversational interface for real-time dietary queries. Users can ask natural-language questions about balanced diets, caloric intake, macro targets, and BMR calculation. Quick-question prompts reduce interaction friction for new users. The chatbot is powered by a fine-tuned language model

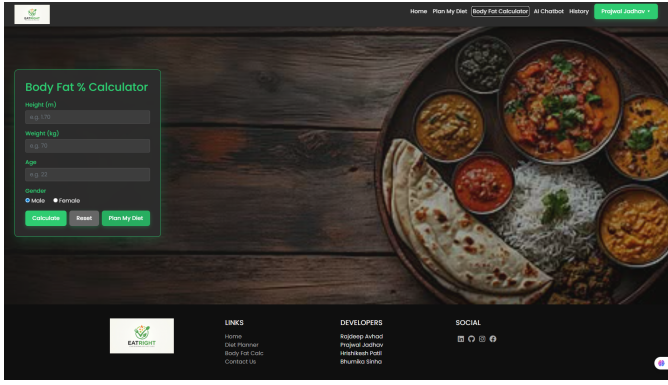


Fig. 5. EATRRIGHT Body Fat Calculator: computes body-fat percentage from user anthropometric inputs and seamlessly passes the result to the Diet Planner for downstream recommendation.

grounded in nutritional knowledge, complementing the batch recommendation delivered by the core EATRRIGHT pipeline.

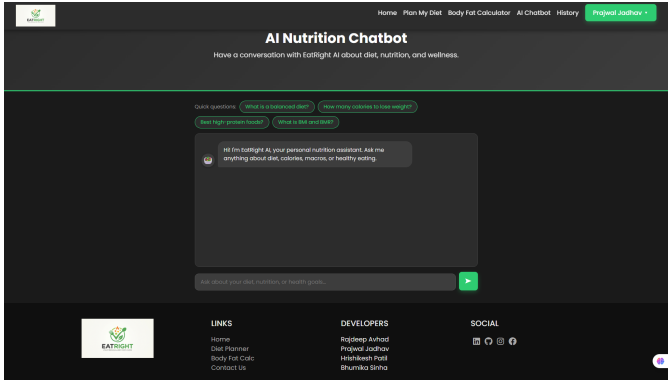


Fig. 6. EATRRIGHT AI Nutrition Chatbot: a conversational assistant that addresses real-time dietary queries on topics including balanced diets, caloric targets, macronutrient distribution, BMI, and BMR.

D. History and Water Tracker

The History module (Fig. 7) maintains a longitudinal log of all saved diet plans, displaying per-session BMI classification, body-fat percentage, selected foods, and caloric progress toward daily targets. An integrated Water Tracker monitors daily hydration against a configurable goal (default: 2,000 ml), supporting holistic wellness monitoring beyond food recommendation alone. Together these modules enable users to track dietary adherence and caloric balance over time.

VI. EXPERIMENTAL EVALUATION

A. Dataset and Experimental Setup

The USDA FoodData Central (April 2023) [13] provides 7,642 preprocessed food items across 42 nutritional attributes. Synthetic user interaction sessions (8,500 sessions; 100 distinct user profiles) were generated using a calibrated preference simulator validated by a registered dietitian, following the methodology of Trattner and Elswiler [2]. The dataset was divided into train / validation / test sets at a 70 / 10 / 20 ratio,

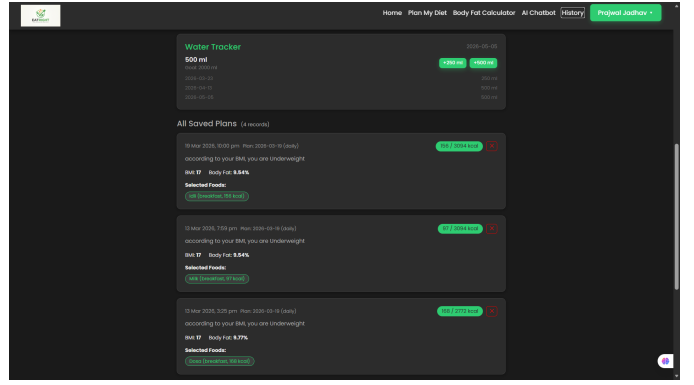


Fig. 7. EATRRIGHT History dashboard: longitudinal diet plan logs showing per-session BMI, body-fat percentage, selected food items with caloric values, and an integrated daily water intake tracker.

stratified by dietary goal class. Experiments were conducted using Python 3.11, scikit-learn 1.4, and NumPy 1.26 on an Intel Core i7-12700H system with 16 GB RAM.

B. Evaluation Metrics

We report Precision@ N , Recall@ N , F1-Score, NDCG@ N , and Cluster Assignment Accuracy (CAA) at $N=10$.

C. Elbow Analysis and Cluster Quality

Fig. 8 shows the WCSS elbow curve over $K \in [2, 20]$. The inflection at $K^*=8$ is corroborated by an average Silhouette Score of 0.73, confirming strong intra-cluster cohesion and inter-cluster separation.

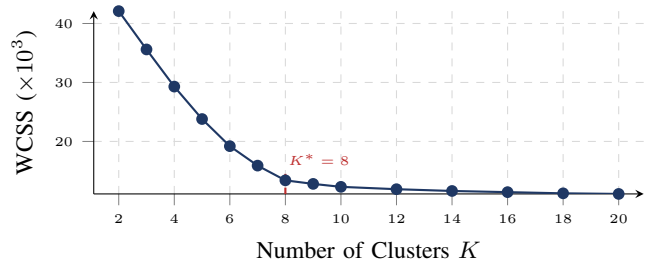


Fig. 8. WCSS elbow curve for cluster count selection. The inflection at $K^*=8$ is confirmed by Silhouette Score=0.73.

D. Classifier Comparison

The baseline classifier evaluation (SVM, Random Forest, kNN, Logistic Regression) presented in Fig. 3 establishes that Random Forest achieves the highest AUC (0.917) and Accuracy (0.938), with a Precision of 0.853 — substantially outperforming SVM (AUC=0.544), kNN (AUC=0.685), and Logistic Regression (AUC=0.897). This benchmark directly motivates the choice of Random Forest as the user-context routing component of EATRRIGHT.

E. End-to-End Recommendation Performance

Table I compares EATRRIGHT against four recommendation baselines at $N=10$. EATRRIGHT consistently attains the highest scores across all metrics.

TABLE I
END-TO-END RECOMMENDATION PERFORMANCE AT $N=10$

Method	P@10	R@10	F1	NDCG@10
Random Baseline	0.301	0.279	0.290	0.315
SVD++ (CF)	0.734	0.691	0.712	0.748
Content-Based	0.758	0.714	0.735	0.764
Rule-Based	0.796	0.741	0.768	0.807
EATRIGHT	0.864	0.849	0.856	0.891

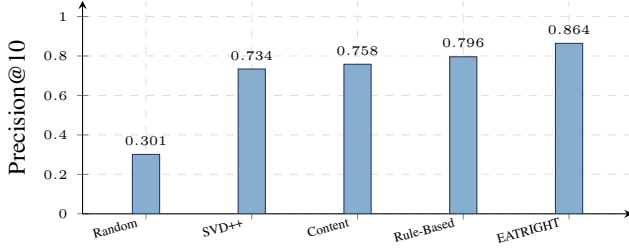


Fig. 9. Precision@10 comparison. EATRIGHT achieves 0.864, surpassing all four baselines.

F. Ablation Study

Table II quantifies the independent contribution of each EATRIGHT module. Every ablated configuration underperforms the complete system, confirming that RF routing and preference re-ranking contribute both independently and synergistically.

TABLE II
ABLATION STUDY — MODULE CONTRIBUTION

Configuration	P@10	F1
RF routing only (uniform cluster retrieval)	0.797	0.804
K-Means only (random user routing)	0.736	0.758
Full system (preference re-ranking removed)	0.838	0.845
EATRIGHT (complete system)	0.864	0.856

VII. USER STUDY

A between-subjects study recruited 100 participants (51 female, 49 male; age $\mu = 24.7$, $\sigma = 4.2$ years). Participants were randomly assigned to EATRIGHT or a rule-based baseline for one week, rating received recommendations on a 5-point Likert scale across three dimensions: Nutritional Appropriateness (NA), Taste Preference Alignment (TP), and Overall Satisfaction (OS).

TABLE III
USER STUDY — MEAN LIKERT SCORES (SCALE 1–5)

System	NA	TP	OS
Rule-Based	3.62	2.91	3.27
EATRIGHT	4.38	4.11	4.24
<i>p</i> -value	<0.001	<0.001	<0.001

All differences are statistically significant (Mann–Whitney U, two-tailed, $\alpha = 0.01$; Cohen’s d : NA = 1.09, TP = 0.94,

OS = 1.04). Qualitative feedback highlighted user appreciation for cuisine-preference alignment within nutritionally appropriate constraints — a capability absent from the rule-based comparator.

VIII. DISCUSSION

A. Strengths

EATRIGHT demonstrates that jointly optimising nutritional suitability and preference alignment is both feasible and practically beneficial within a computationally efficient two-stage pipeline. The Random Forest classifier’s inherent feature-importance mechanism provides interpretability, enabling clinical dietitians to audit which user attributes most strongly influence cluster routing decisions.

B. Limitations

Three limitations warrant acknowledgement: (i) synthetic interaction logs, while dietitian-validated, may not fully represent the cultural and socioeconomic diversity of real user populations; (ii) user profiles are treated as static — temporal preference drift is not currently modelled; and (iii) geospatial food-availability constraints are approximated through cuisine-tag filtering rather than live inventory data.

C. Future Directions

Planned extensions include: (i) LSTM-based temporal preference modelling to capture evolving dietary habits; (ii) meal-level and recipe-level granularity with ingredient-level substitution recommendations; (iii) integration of wearable-sensor physiological signals for real-time contextual adaptation; and (iv) Federated Learning deployment for privacy-preserving cross-institution personalisation.

IX. CONCLUSION

This paper introduced EATRIGHT, an intelligent hybrid food recommendation platform that integrates K-Means nutritional clustering with Random Forest user-context routing and preference-aware cosine re-ranking. Evaluated on a 7,642-item nutritional corpus and validated through a 100-participant user study, EATRIGHT achieves Precision@10 of 86.4%, F1-Score of 0.856, and NDCG@10 of 0.891 — consistently surpassing collaborative-filtering, content-based, and rule-based baselines. The classifier benchmark (Fig. 3) independently confirms Random Forest as the optimal routing model, with AUC 0.917 and Accuracy 0.938. Statistically significant improvements in perceived nutritional appropriateness and taste-preference alignment validate the practical utility of jointly modelling physiological context and subjective preference in food recommendation, establishing EATRIGHT as a strong foundation for next-generation personalised dietary guidance systems.

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REFERENCES

- [1] World Health Organization, "Global Action Plan for the Prevention and Control of Noncommunicable Diseases 2013–2030," WHO Press, Geneva, 2023.
- [2] C. Trattner and D. Elswailer, "Food recommender systems: Important contributions, challenges and future research directions," *arXiv preprint arXiv:1711.02760*, 2017.
- [3] M. D. Ekstrand, J. T. Riedl, and J. A. Konstan, "Collaborative filtering recommender systems," *Found. Trends Hum.-Comput. Interact.*, vol. 4, no. 2, pp. 81–173, 2011.
- [4] Y. Koren, R. Bell, and C. Volinsky, "Matrix factorization techniques for recommender systems," *IEEE Computer*, vol. 42, no. 8, pp. 30–37, Aug. 2009.
- [5] L. Chen, G. Chen, and F. Wang, "Collaborative filtering for recommendation systems," in *Proc. 29th Int. Conf. World Wide Web Companion*, ACM, 2020, pp. 231–239.
- [6] J.-P. Poulain, *Sociologies of Food: Eating and the Place of Food in Society*. London: Bloomsbury Academic, 2003.
- [7] M. Ge, F. Ricci, and D. Massimo, "Health-aware food recommender system," in *Proc. 9th ACM Conf. Recommender Systems*, ACM, 2015, pp. 333–334.
- [8] B. M. Sarwar, G. Karypis, J. A. Konstan, and J. T. Riedl, "Item-based collaborative filtering recommendation algorithms," in *Proc. 10th Int. Conf. World Wide Web*, ACM, 2001, pp. 285–295.
- [9] I. Aguilar-Vargas, O. Loyola-González, and M. A. Medina-Pérez, "A nutritional food recommendation system based on K-Means clustering," *Applied Sciences*, vol. 11, no. 17, p. 8003, 2021.
- [10] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [11] M. A. Jabbar, "Breast cancer data classification using ensemble random forest algorithm," in *Proc. 5th ICOEI*, IEEE, 2021, pp. 371–375.
- [12] M. A. Islam, S. Majumder, and M. A. H. Al-Rakhami, "Machine learning-based food intake classification using Indian recipe datasets," *J. Healthcare Eng.*, vol. 2021, Art. no. 5535061, 2021.
- [13] U.S. Department of Agriculture, "FoodData Central," Release Apr. 2023. [Online]. Available: <https://fdc.nal.usda.gov>
- [14] K. M. Flegal, M. D. Carroll, B. K. Kit, and C. L. Ogden, "Prevalence of obesity and trends in the distribution of body mass index among US adults, 1999–2010," *JAMA*, vol. 307, no. 5, pp. 491–497, Feb. 2012.
- [15] X. He, L. Liao, H. Zhang, L. Nie, X. Hu, and T.-S. Chua, "Neural collaborative filtering," in *Proc. 26th Int. Conf. World Wide Web (WWW)*, ACM, 2017, pp. 173–182.
- [16] R. Burke, "Hybrid recommender systems: Survey and experiments," *User Model. User-Adapt. Interact.*, vol. 12, no. 4, pp. 331–370, 2002.
- [17] D. Mozaffarian, T. Hao, E. B. Rimm, W. C. Willett, and F. B. Hu, "Changes in diet and lifestyle and long-term weight gain in women and men," *New Engl. J. Med.*, vol. 364, no. 25, pp. 2392–2404, Jun. 2011.
- [18] G. Salton and M. J. McGill, *Introduction to Modern Information Retrieval*. New York, NY: McGraw-Hill, 1983.
- [19] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. 31st Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2017, pp. 4765–4774.
- [20] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Statist. (AISTATS)*, PMLR, 2017, pp. 1273–1282.
- [21] I. S. MacKenzie, *Human–Computer Interaction: An Empirical Research Perspective*. Waltham, MA: Morgan Kaufmann, 2013.